

# Quiz — 2.1.2 Thinking Ahead

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Name: \_\_\_\_\_ Date: \_ **Mark:** \_\_\_\_ / 20

OCR H446 · Component 02 · 2.1.2

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## Section A — Multiple choice (6 marks)

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1. A program performs a binary search on a list of customer IDs. The precondition that must hold before the search runs is that the list is: A. Empty B. Sorted into order C. Stored in a database D. Displayed on screen Your answer: ☐
  2. In an online shop's "place order" routine, which of the following is an **input**? A. The confirmation email sent to the customer B. The customer's chosen quantity and delivery address C. Validating the card number D. The updated stock total saved to the database Your answer: ☐
  3. A weather website recomputes the same regional forecast for thousands of visitors each minute. Storing the computed forecast for quick reuse is an example of: A. A precondition B. Decomposition C. Caching D. A race condition Your answer: ☐
  4. Which is the main trade-off a developer must accept when adding caching to the weather site? A. Cached data may become stale and serve out-of-date forecasts B. The site can never use the cache again C. Caching always makes the site slower D. Caching removes the need for inputs Your answer: ☐
  5. A studio reuses a pre-written, pre-tested physics module in three different games. A key benefit of using such a reusable component is: A. It guarantees the games run concurrently B. Faster development with fewer bugs because the module is already tested C. It removes the need for any inputs or outputs D. It makes the games' data automatically sorted Your answer: ☐
  6. Which of these is an **output** of a cash machine's "withdraw cash" routine, rather than an input or a process? A. The PIN typed by the user B. The amount requested by the user C. The dispensed banknotes and updated balance D. Checking the balance is sufficient Your answer: ☐
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## Section B — Short answer (9 marks)

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7. A login routine must run before a user can view their account page. State the **precondition** for displaying the account page, and explain what could go wrong if the routine ran without checking it. [2]

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8. A flight-booking website caches its currency-conversion rates. Discuss **one** benefit and **one** trade-off of caching the rates in this scenario. [2]

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9. For a "calculate total price" routine in a shopping basket, identify **two** inputs and **one** output. Explain why "apply the discount" is *not* counted as an input or output. [3]

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10. A company is building three apps that all need a date-validation feature. Explain **two** benefits of writing this as a single reusable component rather than rewriting it each time. [2]

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## Section C — Applied question (5 marks)

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12. A developer is planning a routine that displays a personalised news feed when a user opens a social-media app. Before writing any code, describe how *thinking ahead* would shape the design. Refer to: the routine's **inputs and outputs**, a **precondition** that must hold first, what data could usefully be **cached**, and one **reusable component** that could be used. [5]

